A Multiband Photometric Re-assessment of Proposed Planet-driven Stellar Pulsations in HAT-P-2

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Abstract

Hot Jupiters on eccentric orbits provide powerful tests of tidal star–planet interactions. In HAT-P-2, Spitzer [de Wit et al., 2017] and HST [Jacobs et al., 2024] analyses have reported stellar pulsations at exact harmonics of the planet's orbital frequency, but stellar models and tidal excitation calculations cannot reproduce these signals, raising doubts about whether they are truly planet-induced or instead instrumental in origin. This project performs the first joint, multi-wavelength reassessment of HAT-P-2 using archival TESS, HST, and Spitzer/IRAC photometry, together with HIRES and HARPS-N radial velocities. If a signal is present only in the infrared but absent in optical and near-UV bands where the F-type star emits most of its flux, this would challenge simple heartbeat-style tidal models or point to wavelength-dependent systematics. If all three bands yield consistent non-detections, the results will strongly disfavor the claimed planet-driven pulsations and bring observations into agreement with current tidal theory.

1 Introduction

Hot Jupiters on eccentric orbits, such as HAT-P-2b ($\sim 9M_J, e \approx 0.5$), are expected to probe the same tidal physics that operate in eccentric "heartbeat" binaries, where time-variable tidal forcing at periastron can excite stellar oscillations at exact harmonics of the orbital frequency [Fuller, 2017]. Separate analyses of Spitzer [de Wit et al., 2017] and HST [Jacobs et al., 2024] photometry of HAT-P-2 have reported candidate pulsations at integer multiples of the planetary orbital frequency, suggesting a possible planetary analogue of heartbeat stars. If these signals are genuinely tidally excited modes, they would provide the first clear example of planet-driven stellar pulsations and a stringent test of tidal excitation and dissipation theories in the extreme mass-ratio regime. However, stellar structure models and tidal excitation calculations to date do not reproduce the reported amplitudes or frequency content, leaving open the possibility that the apparent pulsations are instead produced by unmodeled instrumental, reduction systematics or incompleteness in theory.

A decisive way to discriminate between these scenarios is to test whether the purported signals are coherent and consistent across wavelength. This project conducts the first joint, multi-wavelength reassessment of HAT-P-2 using archival Spitzer (IR), HST (near-UV/optical), and newly analyzed TESS (optical) light curves to test the spectral and temporal coherence of the claimed pulsations. By comparing the presence, phase, and amplitude of signals at the reported orbital harmonics across these datasets, the analysis will determine whether the HAT-P-2 system truly hosts planet-induced heartbeat-like oscillations or whether previous detections are more plausibly attributed to instrument-specific artifacts, thereby constraining both tidal theory and the reliability of single-instrument claims.

2 Proposed Work

Methods and Analysis

This investigation uses archival photometric time series of HAT-P-2 from TESS, HST, and Spitzer/IRAC, together with radial velocity measurements from HIRES and HARPS-N. The TESS and HST light curves have already been modeled and analyzed in detail to test for the presence of the previously reported pulsations at the 79th and 91st harmonics of the planetary orbital frequency. For each dataset, we fit the raw flux with a joint transit and orbital phase-curve model using the batman transit code coupled with an affine-invariant Markov chain Monte Carlo sampler (emcee), allowing key photometric and orbital parameters to vary while fixing others to literature values where appropriate. The best-fitting posterior median model is then phase-folded on the planetary orbital period and subtracted from the raw data, yielding residual light curves in which planetary and geometric effects are removed to first order, isolating any remaining stellar variability.

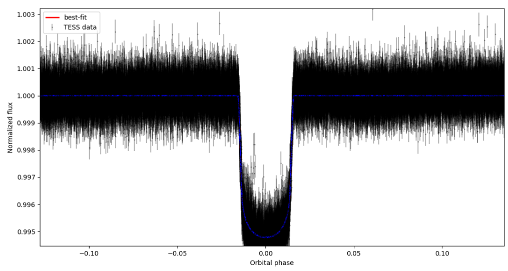

Figure 1: Raw TESS data + best-fit model folded over the orbital period.

We apply generalized Lomb-Scargle periodogram analysis to these residuals and examine the power spectrum in the vicinity of the claimed pulsation frequencies at $79f_{orb}$ and $91f_{orb}$. In both the TESS and HST residuals, no significant peaks are detected at these harmonics above standard false-alarm probability thresholds, including a conservative 5σ false-positive line. To test the sensitivity of our procedure and verify that the non-detections are not driven by analysis choices, we inject synthetic sinusoidal signals at the reported frequencies with amplitudes comparable to those claimed in the literature, re-run the full modeling and subtraction pipeline, and re-compute the periodograms. In these injection–recovery experiments, the artificial signals are robustly recovered above the detection thresholds, demonstrating that our pipeline would have detected pulsations of the reported strength if they were present. Taken together, these results indicate that, in the TESS and HST datasets analyzed to date, we do not find evidence for planet-induced stellar pulsations at the claimed orbital harmonics. In parallel, we have initiated a radial velocity analysis using archival HIRES and HARPS-N measurements of HAT-P-2 [de Beurs et al., 2023]. We model the observed velocities with a single-planet Keplerian corresponding to HAT-P-2b, fixing the orbital eccentricity and argument of periastron to literature values [de Beurs et al., 2023] and allowing independent systemic velocity (γ) and jitter terms for the HIRES and HARPS-N instruments. This approach accounts for instrument-specific zero points and excess noise while focusing on the dominant planetary signal; the putative outer companion HAT-P-2c [de Beurs et al., 2023] is neglected at this stage. The resulting best-fit model provides a baseline against which to search for additional RV structure at harmonics of the orbital frequency that could be associated with stellar oscillations, and it will be extended in subsequent work to incorporate the full multi-planet and multi-wavelength picture.

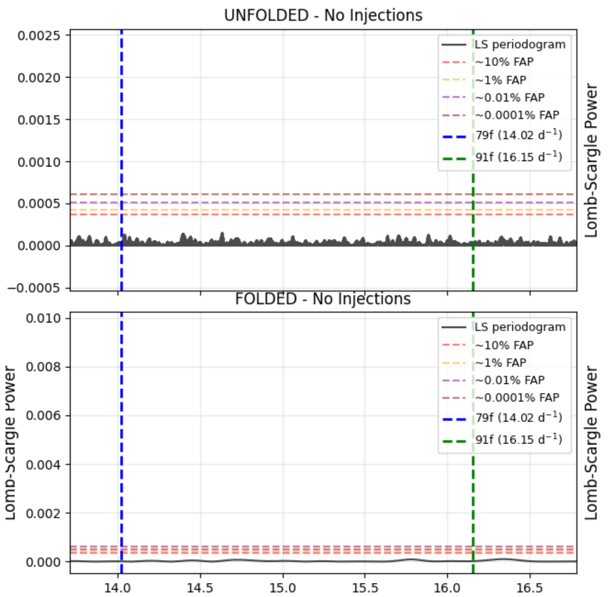

Figure 2: Uninjected LS periodogram of TESS data. We see no real pulsations above the FPLs at the proposed frequencies.

Expected Outcomes

The summer research phase will extend the current analysis in two targeted ways. First, the radial velocity data from HIRES and HARPS-N will be fully integrated into a one-planet Keplerian model for HAT-P-2b. Given the high eccentricity and the resulting short temporal separation between primary and secondary eclipses, this refined RV solution will provide a tightly constrained orbital geometry and phase reference that can be directly linked to the timing of any putative pulsation signals, ensuring that frequency searches at harmonics of the orbital period are placed on a secure dynamical footing. Second, the same analyses already applied to the TESS and HST photometry will be carried out on the Spitzer/IRAC light curves, producing a homogeneous, three-band assessment of the claimed signals.

These steps will yield a decisive multi-wavelength test of the earlier claims of planet-induced pulsations. If statistically significant peaks at the reported harmonics are recovered in the Spitzer infrared data but remain absent in the TESS optical and HST near-UV bands, the result would be physically puzzling: an F-type star radiates the bulk of its flux in the optical, so genuine photospheric pulsations are expected to be at least as prominent in optical light as in the infrared. Such a wavelength dependence would be difficult to reconcile with simple heartbeat-style tidal excitation models and would instead point either to missing physics in current theories or to subtle, wavelength-dependent instrumental systematics in the Spitzer observations. Conversely, if the Spitzer analysis likewise yields non-detections at these harmonics, then the combined TESS+HST+Spitzer data will strongly rule out planet-induced pulsations at the previously reported amplitudes, bringing observations into alignment with existing tidal excitation calculations and suggesting that the original signals arose from analysis or instrument systematics rather than true stellar oscillations. In either case—multi-band inconsistency that challenges current theory or a coherent null result that supports it—the outcome will be a clear, publication-quality statement about the physical reality of the proposed pulsations in HAT-P-2 and the reliability of prior single-instrument claims.

3 Mentorship Component

Dr. John J. Zanazzi, whose research centers on tidal dynamics and exoplanet architecture, mentors this project through regular weekly in-person meetings during the academic year, where we review analysis decisions, interpret results, and set next steps. His guidance is explicitly focused on building my independence: I design and implement the photometric and radial-velocity analysis, while he challenges the physical interpretation and connects the work to current theory and literature. Because the project is entirely computational and uses archival data, it is well suited to a remote format over the summer; Dr. Zanazzi has prior experience supervising and publishing with remote students, and we will continue with scheduled weekly video meetings and shared code/manuscript development to maintain the same level of supervision and academic rigor.

4 Conclusion and Future Directions

Summer funding would make it possible to complete the remaining analysis and move the project to a publication-ready stage within a single, focused period, rather than spreading the work thinly across future semesters. I will be working from India over the summer, but the analysis pipeline is already established on my own computing environment, and the research does not depend on laboratory facilities or on-site observing. Regular remote meetings with Dr. Zanazzi and shared code/manuscript workspaces will extend the effective mentoring structure that has been in place during the academic year, so that the change in location does not interrupt progress.

Looking beyond the immediate Spitzer and radial-velocity analysis, a natural next step is to develop a theoretical counterpart using stellar evolution and oscillation modeling with MESA [Fuller, 2017]. Future work will model tidal excitation with MESA [Fuller, 2017], comparing predicted pulsation spectra to our observational limits and extending to other eccentric hot Jupiters, forming the basis of a longer-term research program.

References

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